Methods

Selection of community forest sample

We used a stratified random approach for selecting CFAs for the survey. First, we identified eight provinces for sampling, omitting Luapula and Copperbelt provinces which had few CFAs. Our intention was to select ten CFAs per province stratified by size and age (preference for older CFMAs). However, we found that three of the selected provinces had fewer than ten qualifying CFAs. Hence, to maintain the number of CFAs at 80, we selected 12 CFAs from provinces that had more than ten. We established a cut off establishment date of 1st Jan 2024, so that all selected CFAs were at least 1 yr old. In addition, we were not able to use CFA size as a selection criteria, because of the low number CFAs in several of the provinces. Hence, within provinces, we simply selected our sample randomly from among the CFAs that were at least 1 yr old.

Survey tool

We developed a survey questionnaire tool to assess the quality of CFM implementation and outcomes for SFM and sustainable development. The same tool was used in the FGDs and in the individual interviewees. The structured design enabled comparisons among the responses from different FGDs, individual interviewees and across CFAs. However, interviewees were encouraged to discuss the appropriate response, and comments sections throughout the tool enabled free text responses that could capture information not listed among the options in the tool. The complete tool is provided in Appendix II. The tool was fully pre-tested at M'phansha CFA in Rufunsa and feedback from the survey team, the FDG groups and the individual interviewees was incorporated into the final version of the tool.

The tool covered CFA (i) governance including business activities, (ii) SFM outcomes, such as impacts on deforestation, fire frequency, and wildlife, (iii) economic transformation including the impact of CFM on access to forest resources, employment opportunities, income earning opportunities and opportunity costs, (iv) equity and inclusion, including benefit sharing arrangements and their impact on vulnerable groups, and (v) opportunities for improving CFM implementation.

In each CFA forest, the team conducted four FGDs including (i) CFMG executive, and community (ii) females, (iii) males and (iv) youth. Thus, we are able to compare responses among these stakeholder groups. In addition, the survey team conducted individual interviews with key stakeholders. The aim was to interview (i) a CFMG executive member, usually the secretary or chairperson, (ii) the DFO, (iii) traditional leadership, usually a village headperson, (iv) the chairperson of a forest user-group, and (v) a representative of any external organisation supporting the CFM process, such as a conservation NGO or a carbon developer. Again, by consistently selecting interviewees from among these different types of key stakeholder, we are able to compare responses within and across CFAs. However, the number of FGDs and individual interviews were were able to conduct varied according to the availability of the stakeholders when were were conducting the survey (Appendix I).

Survey of community forests

The surveys were carried out according to the following schedule in Table 1.

Table 1: **Table 1. CFA survey schedule**

Province	Dates
Lusaka	13 March to 17 April 2025
Southern	8 May to 28 May 2025
Central	8 June to 21 June 2025
Eastern	8 July to 31 July 2025
Northern	10 Aug to 2 September 2025
Muchinga	10 September to 5 October
North-western	17 October to 11 November 2025
Western	26 November to 8 December 2025

During the survey, five CFAs were dropped from the sample (Table 2). For various reasons these CFAs existed on paper, but were non-functioning on the ground.

Table 2: **Table 2. CFAs selected but not surveyed**

Province	CFA	Reasons for omission
Lusaka	Chalimbana	The CFA was not operational and currently exists only in the database. Only the initial establishment stages were completed.

Interviewer	cfmg_executive	cfmg_female	cfmg_male	cfmg_youth	Individual interviews
Chipo Chisonga	46	1	47	0	64
Edward Banda	23	1	22	51	104
Hockings Mambwe	3	0	0	9	10
Hope Chame	1	12	0	0	9
Memory Sakuwaha	0	1	0	0	10
Ronald Chamwaza	0	0	0	1	0
Thecla Masuku	2	61	3	8	60

Province	CFA	Reasons for omission
Central	Kaindu	The Chief had disbanded the executive committee and hence it was not possible to organise the community
Muchinga	Nampando	The CFA was not operational and currently exists only in the database. Only the initial establishment stages were completed.
Eastern	Kalumphila	This CFA has not been functional since the partner (Land Alliance) withdrew in 2022. The CFAMG establishment processes were not concluded.
Eastern	Koloko	This CFA has not been functional since 2022 after the partner (Land Alliance) withdrew.

The final list of CFAs surveyed is given in Appendix II.

The surveys were conducted by the following interviewers (Table 3). Chipo Chisonga (Landscape Alliance), Edward Banda (Landscape Alliance) and Thecla Masuku (Forestry Department) comprised the principle survey team with others supporting them as required. Thecla Masuku, supported by Hope Chame (Forestry Department), were responsible for conducting the community female FGDs and female individual interviewees. Given the gender norms in local communities, it was felt that female interviewees might feel inhibited to answer honestly with male interviewers.

: Table 3: Number of interviews conducted by each interviewer, by interview type.

Analysis of the survey results

Data from the survey tool were summarised and displayed using tables, bar charts or word clouds for informative interpretation. To examine hypotheses concerning differences in responses among FGD types or interviewee roles for the individual interviews, we used Generalised Linear Mixed Models (GLMM). For binary responses (e.g. yes/no), we used binomial models and for ordered categories (e.g. decrease, no change, moderate increase, significant increase), we used Cumulative Link Mixed Models (CLMM). Reflecting the nested structure of the data with interviews nested within CFAs nested within provinces, we implemented random intercepts for province and CFA name nested within province (i.e. province:CFA name interaction). For the FGD responses, FGD types was included in the model as a factor and we examined the pairwise differences among FGD types using Tukey Honest Significant Differences (HSDs). For the models investigating interview roles for the individual interviews, we inserted interviewee gender, interviewee age and interviewee role as factors in the models and the pairwise differences among interviewee roles were examined using Tukey HSDs. Some models also included governance index as a continuous explanetory variable.

To provide an overall summary of CFAMG performance across the four thematic areas, we constructed four composite indexes. The **governance index** (0–10) counts the number of ten governance features reported as present by the CFAMG executive FGD: receipt book, bank account, financial reporting, annual reporting, issuing permits, HFOs appointed, workplan, budget, business plan, and term limits. The **SFM index** (0–5) sums five binary indicators averaged across all four FGD types: reduced forest clearing, reduced or absent charcoal production, reduced fire incidence, increased wildlife, and improved forest resource management. The **economic transformation index** (0–5) sums five binary indicators averaged across all four FGD types: improved household access to natural resources, improved access to natural resources for sale, new formal employment opportunities, collaboration with a private enterprise, and absence of opportunity costs (scored 0 if opportunity costs were reported, 1 if not). The **equity and inclusion index** (0–5) sums five binary indicators averaged across all four FGD types: equitable benefit sharing, women benefiting from CFAM, women participating in decision making, transparency in benefit sharing, and existence of a grievance and redress mechanism. For each index, linear mixed models with FGD type and the governance index as predictors and random intercepts for province and CFA within province were used to test whether governance and FGD type were significant predictors of performance. Mean scores for all four indexes by CFAMG are provided in Appendix IV.

Underlying assumptions

This study rests on several key assumptions. First, we assume that CFMG executives are reasonably knowledgeable about governance processes, financial management practices, and CFMG policies, and that community members have a reasonable understanding of forest conditions and livelihood changes in their CFA. Second, we assume that the stratified random sample of 75 CFMGs is representative enough to allow for national-level inferences about CFAM implementation in Zambia, based on geographic coverage across eight provinces and a range of CFA sizes and ages (see Survey chapter). Third, we assume that the multi-perspective FGD design (separately interviewing executives, women, men, and youth) will capture genuine variation in perspectives across community stakeholder groups, rather than systematic biases tied to these categories. Fourth, we assume that respondents can and will recall relevant events and changes over the 1-7 year history of their CFMGs with reasonable accuracy. Fifth, we assume that the composite indexes (governance, SFM, economic transformation, equity & inclusion) appropriately capture the underlying constructs they purport to measure, and that aggregating binary indicators into indexes preserves meaningful information about CFMG performance.

Statistical assumptions

The statistical models employed make standard assumptions. Linear mixed models assume normality of residuals and homogeneity of variance across groups. Cumulative Link Mixed Models (CLMMs) assume that the proportional odds assumption holds—that the effect of predictor variables is consistent across outcome categories. Generalised Linear Mixed Models assume that the link function appropriately transforms the response variable and that the variance structure is correctly specified. However, with reasonably large sample sizes (75 CFMGs, 292 FGDs, 257 individual interviews), the models are relatively robust to moderate violations of these assumptions, particularly for the fixed effects estimates. Random intercepts for province and CFA within province account for the nested structure of the data, but we have not tested for random slopes, which might be important if the effect of predictors varies substantially across provinces - although this is unlikely. The governance index, treated as a continuous predictor in models, is actually an ordinal count variable (0-10), which could affect interpretation if its effect is non-linear.

Potential response biases

Several response biases could affect the survey results. **Social desirability bias** is the risk that respondents report what they want interviewers to hear rather than their true experience. This may be particularly acute for sensitive topics like elite capture, financial mismanagement, or inequitable benefit-sharing. We attempt to mitigate this through the multi-perspective design: CFMG executives may downplay governance problems, but community groups are more likely to report them. **Knowledge asymmetry** is a related concern: CFMG executives have more information about formal governance processes, while community members may be less informed, but may perceive outcomes more directly. **Interviewer effects** could introduce bias if individual interviewers systematically obtain different response patterns. However, preliminary analyses suggested there was no significant interviewer effect. **Seasonal variation** is another concern: responses about employment and income may differ depending on whether the survey was conducted during a high or low economic season in that community. The survey spanned March-December 2025, which includes both wet and dry seasons in Zambia, potentially introducing seasonal variation in perceived livelihood impacts. **Recall bias** may affect responses about historical changes in forest condition or employment: respondents recalling changes over 1-7 years may misremember, conflate recent changes with longer-term trends, or be influenced by current conditions.

Questionnaire limitations

While the survey tool was pre-tested and refined, several limitations remain. **Comprehensibility**: The questionnaire was administered in English, Bemba, Nyanja, and Tonga depending on the region, but the translation and interpretation of technical terms (e.g., "equitable benefit-sharing," "sustainable forest management," "governance") could vary. Respondents may have interpreted questions differently than intended, particularly questions requiring numerical estimates (e.g., "What percentage of community members participate in CFMG decisions?"). **Response options**: The questionnaire used fixed response categories (e.g., yes/no, significant/moderate/no change) which may not capture the full complexity of respondents' experiences. Respondents forced into categorical responses may have felt their nuanced view was not represented. **Coverage**: The questionnaire covered five main domains (governance, SFM, economic transformation, equity & inclusion, improvements) but could not address all aspects of CFM comprehensively. For example, questions on dispute resolution, grievance mechanisms, and conflict management were limited compared to governance and forest management topics. **Question framing**: The order of questions could introduce framing effects—early questions may influence how respondents answer subsequent questions. The questionnaire also included open-ended comment sections, which captured important qualitative information but resulted in variable depth of response depending on respondent verbosity and interviewer probing effort. Finally, the questionnaire relied heavily on self-report of practices and conditions; we have no independent verification of whether reported governance practices are actually implemented (except through the remote sensing analysis for forest condition claims).

Remote sensing analysis

Data sources and metrics

For the remote sensing analysis we used MODIS active fire detection data (fire detection density, burned area, fire radiative power), the Hansen Global Forest Change 2024 tree cover loss data, and ALOS-2 PALSAR-2 ScanSAR above-ground biomass estimates for 2018–2024 (Harrison et al. in prep).

Spatial and temporal scope

We included all CFAs with boundary information which included 40 CFAs our of the 75 in the survey (53% of surveyed CFAs). Temporal coverage included 2014-2025 and we included 1 km buffer zones for leakage assessment.

Analysis method

We employed Linear mixed models with random intercepts for CFMG and used a Before-After comparison and a Core vs. Buffer design. Any significant After:Buffer interaction would have indicated a leakage effect. Governance index was included as a fixed covariate. Linear mixed models assume normality of residuals and homogeneity of variance. We performed the normal model tests on the residuals to ensure they performed satisfactorily.